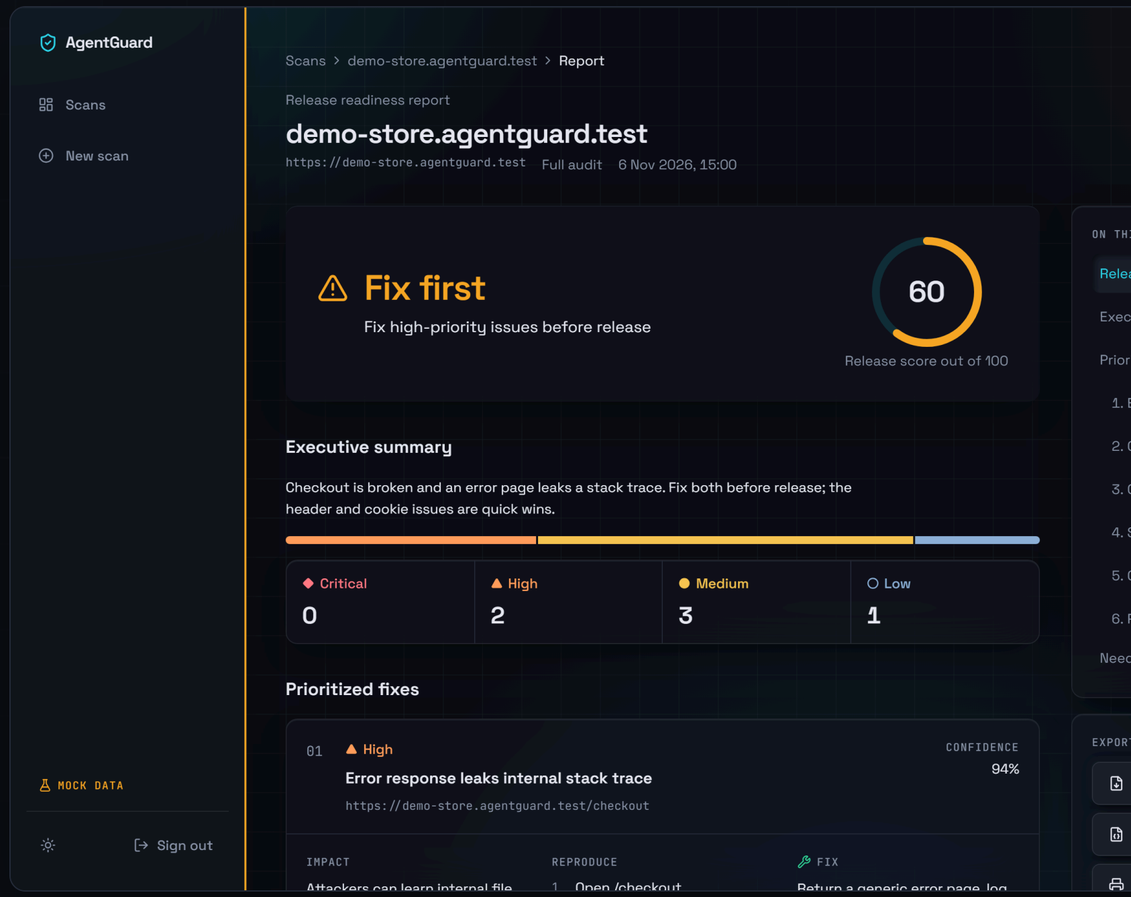
AgentGuard

Tests your web app before your users do.

An autonomous AI security and QA engineer. Real browser tests in disposable sandboxes, evidence for every finding, and a clear release decision.

Ayesha, **Sohaib**, **Sarah** and **Arslan** for the LabLab hackathon



UI shown with demo data

Releases ship with problems nobody checked.

Broken checkout

A valid order returns an HTTP 500 and the sale is gone.

Leaked stack traces

Error pages show file paths and framework versions to anyone.

Weak security headers

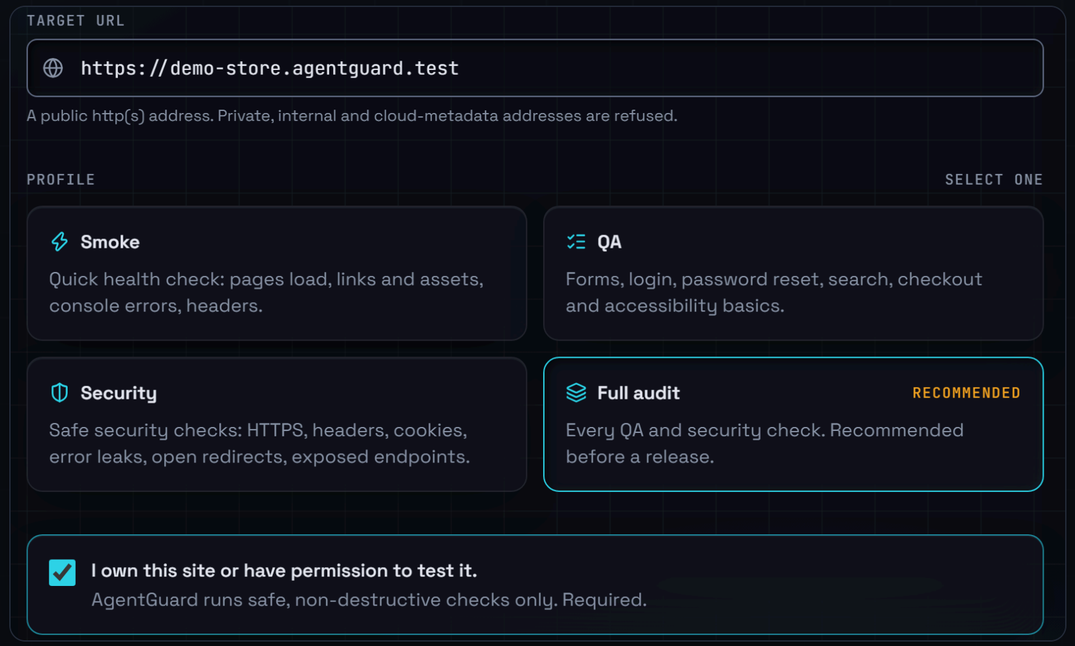
No CSP, no HSTS, cookies without Secure or HttpOnly.

Inaccessible pages

Unlabeled inputs and images without alt text.

QA and security reviews are **slow, manual and run by different people**, so these slip through right before a release.

A URL in. A release decision out.



1 Enter a URL, pick a profile, confirm you are allowed to test it.



2 Get a score, a verdict and the fixes, each backed by evidence.

UI shown with demo data

One clear decision, backed by a score.



UI shown with demo data

Block release	0-49
Fix first	50-79
Safe to ship	80-100

Start at 100 and subtract per finding:
`critical -25, high -12`
`medium -5, low -1`

Plan, run, verify, decide.



URL + profile

Smoke, QA, Security or Full audit

AI plan

Picks checks from a whitelist of 18

AI STEP

Sandboxed browser tests

Playwright, one container per task

Verifier

Re-checks findings against saved evidence

AI STEP

Score

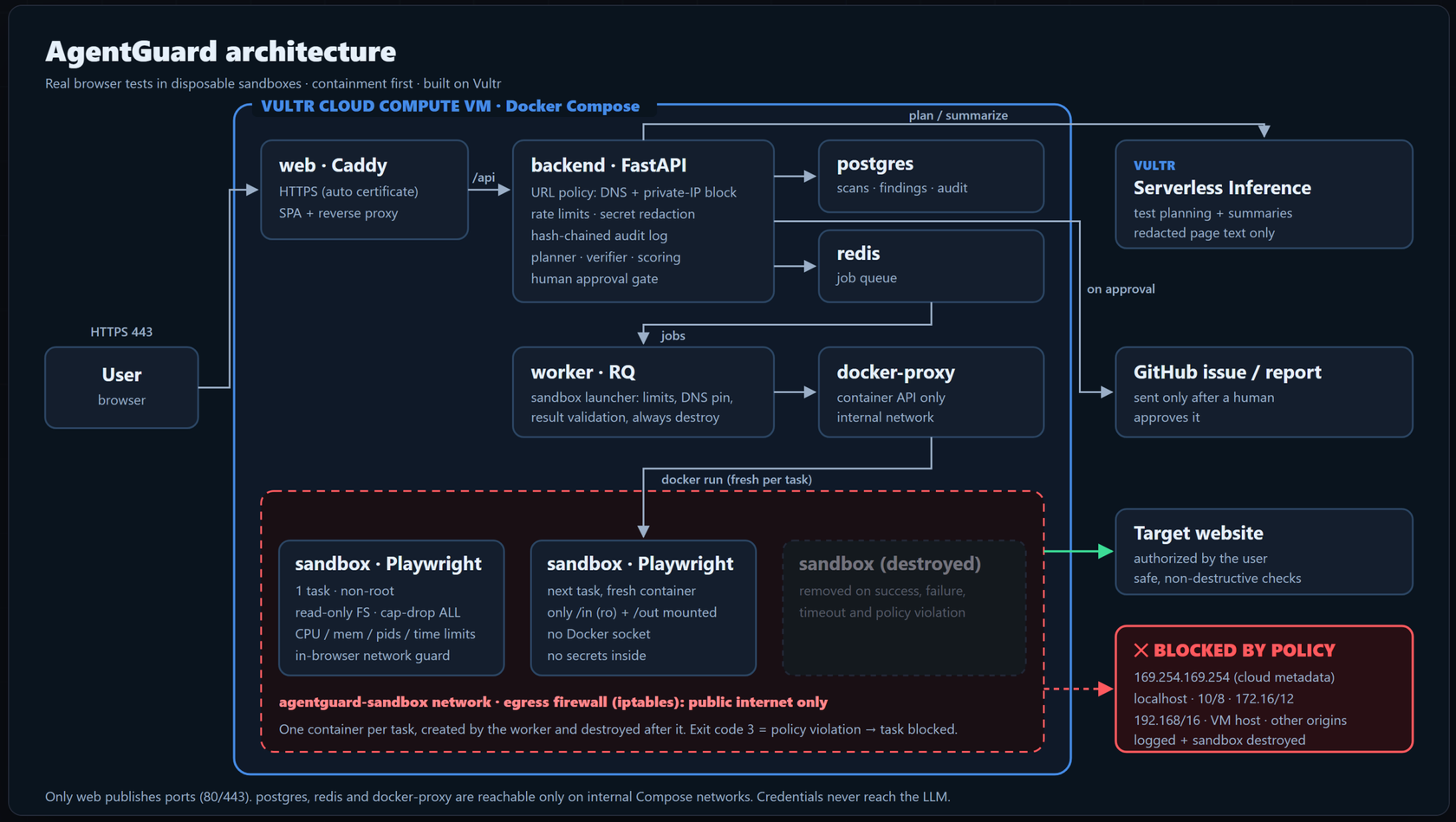
0 to 100 and a verdict

Human approval

The decision is recorded in the audit log

- The planner can only choose from 18 whitelisted checks. It can’t invent new ones, and a fallback plan runs if the model is down.
- Every test runs in a real browser inside a throwaway sandbox.
- The verifier re-checks each finding against saved evidence before it counts.

One VM, one sandbox per test.



- Web app, API, queue and Postgres in Docker Compose on one VM
- Serverless Inference plans each scan
- Every test runs in a fresh Playwright container, deleted afterwards

Status. Runs locally today; the Vultr deployment is in progress. Approvals are recorded only: nothing is sent to GitHub or by email yet.

Designed for Vultr.

Cloud Compute VM

Hosts the app, API, database, queue and every sandbox. Owning Docker and the host firewall is what lets us cut each sandbox off at the network level.

Serverless Inference

Test planning and summaries over an OpenAI-compatible API. No GPUs to manage.

Firewall

Only 80/443, plus SSH from the team, can reach the VM.

Deployment in progress. The stack runs locally today. The inference client is tested with a scripted model, not yet against the live Vultr endpoint; if the model fails, the built-in fallback plan runs.

Containment first. We assume the site is hostile.



The containment demo task tries to reach the cloud metadata address. UI shown with demo data.

- One container per task, always destroyed
- CPU, memory, process and time limits; non-root
- No Docker socket, no host files, no secrets inside
- Private IPs, localhost and cloud metadata blocked in 3 layers
- Credentials never reach the LLM; secrets are redacted
- Tamper-evident, hash-chained audit log

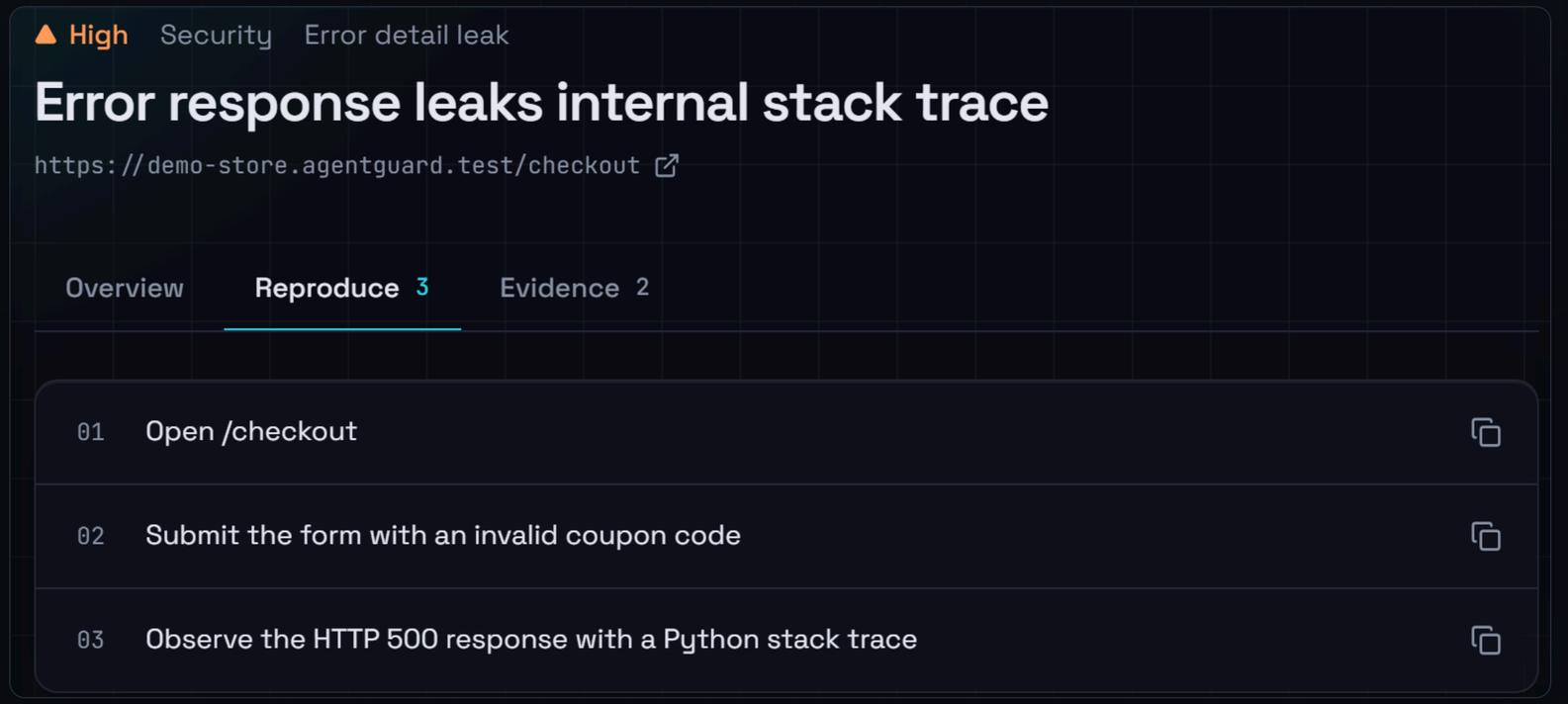
The AI decides what to test, not what is true.

	BEFORE VERIFIER	WITH VERIFIER
Real findings re-checked against evidence	0 of 13	13 of 13
Real findings lost	0	0
False alarms that reached the score	6 of 6	0 of 6

Real check output from our demo store plus 6 injected faults, no LLM in the eval. Method: docs/agent-eval.md

- **Planner:** whitelisted checks, task caps, one repair round, deterministic fallback
- **Verifier:** evidence re-check, dedupe, severity guide; unbacked claims never count
- **Prompt injection:** site text is redacted and fenced as data, and every answer is validated

Every finding comes with proof and a fix.



UI shown with demo data, modelled on our deliberately flawed demo store.

Severity + confidence
Clamped to a severity guide

Reproduction steps
Exact steps to see it yourself

Evidence
Screenshot and network summary from the sandbox

Suggested fix
What to change before release

Who needs it, and what comes next.

Who it is for

Product, QA and security teams

One release check instead of separate, manual reviews

Agencies

Evidence-backed reports before handing a site to a client

Startups before launch

A safety net without a dedicated security hire

Roadmap

FUTURE WORK, NOT IN THE DEMO

CI/CD integration
Scan every pull request and preview deploy

Authenticated flows
Test pages behind a login

Scheduled scans and more profiles

Send approved GitHub issues and PDF reports
Today approvals are recorded only

The team

Ayesha

UI, DOCS AND DEMO

Web app and reports

Sohaib

BACKEND

API and orchestration, QA checks, demo store

Sarah

AI / AGENT

Planner, verifier and scoring

Arslan

INFRA + SECURITY

Sandbox, security checks, policy and audit, Vultr deploy



Code on GitHub

Scan to open the repository

github.com/arslan-ahm/AgentGuard-The-Redteam-Agent-lablab.ai-project-

Demo video

Coming soon

Live demo

Coming soon, deployment in progress